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Table of Contents:

Cover

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A Short Editorial

Question I: Why do we not know what dark matter is?

Question II: If bones are four times as strong as concrete, why do they break so easily?

Question III: What If Your Brain Is Just Predicting Reality Instead of Seeing It?

Question IV: Why are some non-metals so strange?

Question V: How do planes stay in the air?

A Brief Newsflash

Acknowledgements

Conclusion

A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: Why do we not know what dark matter is?

– Danyar Mehrad, 17

The Universe is the vast vacuum of space we live in, the infinite expanse of galaxies, constellations, and nebulae, the unlimited space our imaginations can barely grasp.

We are made up of matter. Everything is. The fly buzzing around in the room, you reading this article, and the planet we live on. Matter makes up all of that. So, since matter makes up everything, it must make up all of the universe, right?

Not exactly. Matter, visible matter, makes up only 5% of the entire universe. It may seem hard to grasp at first. But this begs the question, what about the other 95%? The other 95% consists of three components: antimatter (trace amounts), dark matter (~27%), and dark energy (~68%).

Antimatter is like matter's evil twin. Matter and antimatter, if ever colliding, would annihilate one another, releasing pure energy. If one gram, one tiny gram, of each were to collide, the destruction could be comparable to a nuclear bomb. The reason anything still exists is that there was some more matter than antimatter by just a bit, so when everything was annihilated, there was more matter in the end. Antimatter was not all destroyed, but a large amount of it was. What remains is considered negligible. We owe our existence to the battle that matter won.

Moving on, dark matter is an invisible version of the matter that makes us up. It does not reflect, absorb, or emit any wavelength of electromagnetic waves. Its importance lies in the fact that without it, our universe's infrastructure would crumble. Dark matter exists to complete matter. It exerts a gravitational force that keeps the astral bodies of our universe stable.

Dark energy is different. No one really understands it. Many scientists have different theories about it. But what is known is that dark energy causes the expansion force of the universe. It combats the inward force of gravity. Lately, dark energy has been acting up. The rate of the universe's expansion is increasing, and we know dark energy is responsible, but we do not know why.

Question II: If bones are four times as strong as concrete, why do they break so easily?

– Satvika Satish, 11

Bones are a main part of the bodies of living creatures, providing the structure that muscles are built on and to support an organism and allow it to move and function. Human bones are strong and withstand pressure enabling us to run around or exercise easily, but how strong are they actually?

Tests show that bone material is approximately 4 times stronger than concrete when a squashing force is applied, where compressive forces are forces causing the object to contract or get squished, concrete is good at handling such forces but not so much other forces such as twisting forces.

Then why do our bones break?

When a person falls they may land on a particular spot on a bone and it may break or fracture, but that depends on which part of the bone takes the impact of the fall, different bones have different shapes, they are built for different tasks, tasks requiring strength in various places of the bone, bones are built to handle pressure in particular directions and areas. A person may fall on a bone such as the femur and it doesn't break, but could fall with the same impact force on a different part of their femur and it breaks, this would happen because that part of the femur is less capable of absorbing force than the part the person fell on the first time.

Another thing would be the strength used to compare bones and concrete. Concrete is a material capable of withstanding squashing forces but is weaker at handling other types of stress like twisting and sliding forces. The thing is, it isn't only how the force is applied, it is also about the impact speed and sudden increase of stress over a small area. The smaller the area of impact the higher the chances of fractures as the force increases extremely fast and the impact speed affecting as well because a higher impact speed is caused by a higher change in momentum therefore the bone is unable to handle the change and breaks. Nonetheless, bones are very useful in many ways and with an extremely powerful structure help us in everyday tasks even when we don’t realize it.

Question III: What If Your Brain Is Just Predicting Reality Instead of Seeing It?

– Noora Abdullah, 13

You trust your eyes, don't you? It feels obvious – what you see must be real. But what if your brain isn't actually showing you the world as it is? What if it's guessing… and you're just going along with it?

Scientists now believe the brain works less like a camera and more like a prediction engine. Instead of passively receiving information, it constantly makes rapid guesses about what's out there, using past experiences, memories, and expectations. By the time your eyes send signals, your brain has already built a version of reality – and simply adjusts it if needed.

That explains why optical illusions are so convincing. A still image seems to move. Two identical colours look completely different. Your brain isn't broken – it's just trying to predict too quickly, filling in gaps with assumptions that don't quite match reality.

This doesn't just happen in strange images. It shapes your everyday life. When you walk into a familiar room, you don't carefully examine every object. Your brain predicts what should be there, saving time and effort. But this shortcut comes with a catch: sometimes, you're not seeing what's actually there – you're seeing what you expect to be there.

Even more unsettling, this means two people can experience the same moment differently. Their brains, shaped by different memories and emotions, construct slightly different realities. So who is right?

Maybe both. Or maybe neither.

The idea is simple, but uncomfortable: reality isn't something you passively observe. It's something your brain actively builds. And if that's true, then the world you trust so completely might not be as solid as it seems. Which begs the question – who CAN you trust?

Question IV: Why are some non-metals so strange?

– Auraka, 17

You are surrounded by countless elements, metals and nonmetals alike. Despite most elements being classified as per their properties, a few stand out for their variable behavior under inordinate conditions.

Hydrogen, is just a gas with a single proton and electron, right? Well, not so much. If given the right amount of heat and pressure, metallic hydrogen – a phase of hydrogen where atoms are forced into a metallic state, allowing electrons to move freely – will form. Metallic hydrogen does not exist naturally on Earth – it only exists deep inside giant planets such as Jupiter and Saturn. On Earth, it has only been created in high-temperature, high-pressure laboratory environments; one successful operation was conducted by Harvard University, lasting for just under six months before the experiment was put to an end.

Nitrogen, unlike hydrogen, is a stable gas. When forced under extreme conditions, the molecules break apart, and nitrogen gains metallic properties, such as thermal stability. One special solid form, Cubic Gauche nitrogen (cg-N), a high-energy-density material (HEDM) that, as the name highlights, stores a huge amount of energy that is released once it decomposes into nitrogen gas. These exotic versions don't last outside high pressure, but they show how versatile nitrogen can be.

Oxygen, on a day-to-day basis, is a familiar diatomic gas that we rely on to live. However, once cooled and compressed, it passes through a series of solid phases with striking colors – red, orange, and deep blue – each reflecting a change in how the O_2 molecules arrange themselves. In some of these dense forms, oxygen becomes electrically conductive, and in its metallic state, it can even act as a superconductor at low temperatures. These transformations don't persist once conditions return to normal, but they reveal how oxygen can shift from a life-sustaining gas into a material with various properties unlike its own.

Question V: How do planes stay in the air?

– *Kamyar Muhammed, 12*

There are a few different forces at work here. There is lift, weight, thrust, and drag. But the biggest force at work here is definitely the lift, since it is the one opposing the plane's weight. But how does this lift actually occur? Well, it is once again the old airplane wing. Planes stay in the air because of this force that is a result of the way in which they are shaped. It is called an airfoil. It is this shape that allows the air to pass over the top of the wing faster than it passes under. This results in a pressure difference since there is more pressure on the bottom than there is on top. It is this way in which we get this lift. It is really rather cool to think about, since it is really a matter of this balance and it is really dependent on this.

Another way to look at lift is through Newton's Third Law of Motion, which states, "For every action, there is an equal and opposite reaction." The wings, in pushing the air downward, experience an upward reaction from the air, which causes the airplane to rise.

However, for the lift to be produced, the airplane has to be in motion. This motion is achieved through the application of thrust, which is usually in the form of either a jet engine or a propeller. The application of the propeller or the jet engine pushes the airplane through the air, and this is what generates the lift.

When a plane is flying through the air, it experiences drag, which always wants to slow down a plane. In order to fight this force, pilots and those who design planes always attempt to create a plane that minimizes drag as much as possible. This way, the engines of the plane will be able to push it forward. If the engines are pushing harder than the drag is pulling back, then the plane will start moving faster. But if the drag and the push of the engines are equal, then the plane will simply continue flying at the same speed.

So, in the case of an airplane, the ability to stay up depends entirely on how hard the engines are working. It's a pretty delicate balance, but when they get it just right, the airplane stays in the air.

A BRIEF NEWSFLASH

NASA is going back to the moon in the beginning of April, for the first time since 1972! This is an event to enjoy, so make sure to check in on the launch and the Vlogs that the astronauts will be making!

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mission to help the Iraqi youth where there
has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at https://YSJOI.is-great.org

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